

HARINARAYANAN U

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Summary

Full-stack engineer who ships across frontend, backend, APIs, and AI-powered workflows. Published an open-source distributed job queue on PyPI, built a production LLM pipeline serving enterprise clients, and developed a React + TypeScript frontend for a multimodal AI application. Comfortable moving fast across the stack, leveraging AI developer tools to accelerate delivery, and taking early ownership in fast-moving engineering environments. Passionate about building scalable platforms that help people work better together.

Education

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.28/10

Nov. 2022 – Sept. 2026 (Expected)

Kollam, Kerala, India

Experience

RevGain AI

Sept. 2025 – Feb. 2026

Software Developer Intern

Hyderabad, India

- Built and shipped a scalable backend API service (Python + FastAPI + PostgreSQL) serving enterprise clients in production on AWS EC2 with Docker — owned architecture, implementation, CI/CD deployment, and iterative improvement end-to-end.
- Designed and integrated an AI-powered workflow — LLM pipeline using RAG, schema-linking, and few-shot prompting to automate enterprise analytical queries, improving accuracy to ~85% (~40% error reduction vs. baseline).
- Leveraged AI developer tools throughout the SDLC to accelerate delivery — debugged production issues, improved application performance, and ensured service reliability under concurrent multi-tenant load (<600ms p95 latency).
- Collaborated with product managers and data teams in Agile sprints — delivered features quickly and iteratively, writing clean, testable, and maintainable code following engineering best practices.

TiVo

Aug. 2024 – Nov. 2024

Software Developer Intern

Bangalore, India

- Debugged production performance issues in AVPlayer — traced root cause through caching and state handling, shipped targeted fix reducing incidents by ~25% at scale.
- Contributed to 50+ code reviews enforcing quality and best practices; delivered 8+ UI component redesigns improving maintainability across the application.

Projects

SONAR | *Python, PyTorch, RoBERTa, openSMILE, FastAPI, React, TypeScript* | [GitHub](#)

Sept. 2025 – Present

- Built a full-stack AI application — TypeScript + React frontend with real-time emotion display and music recommendation UI, backed by a Python FastAPI inference service running multimodal deep learning models (RoBERTa + openSMILE).
- Owned both frontend and backend layers with no handoff — built responsive UI components, REST API integrations, and confidence-aware model fusion end-to-end; published at IEEE IATMSI 2026 ([IEEE Xplore](#)).

QueueCtl | *Python, SQLite WAL, Flask, Socket.IO, Docker* | [GitHub](#)

Nov. 2025 – Present

- Designed and published an open-source distributed job queue on PyPI — atomic job claiming, DAG-based dependency resolution, Dead Letter Queue, exponential backoff retry, HMAC-signed webhooks, and real-time Socket.IO observability dashboard.
- Shipped real-time Flask + Socket.IO observability dashboard; wrote unit and integration tests validating atomicity guarantees and retry behaviour under concurrent load; Dockerized with CI/CD for scalable deployment.

Publications

SONAR: Hybrid Explainable Emotion-Aware Music Recommendation | [IEEE Xplore](#)

Mar. 2026

- IEEE IATMSI 2026 — Multimodal AI system (RoBERTa + openSMILE) with confidence-aware fusion and full XAI pipeline; IEEE Access journal submission in progress.

PrivMod: Federated Learning for Privacy-Preserving Content Moderation | [IEEE Xplore](#)

Dec. 2025

- I4Tech 2025 (Scopus Indexed) — 94% accuracy under differential privacy ($\epsilon = 0.5$) across 100K+ samples using federated CNN-BERT pipeline.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C/C++

Frontend: React, TypeScript, HTML, CSS

Backend & APIs: FastAPI, Django REST, PostgreSQL, SQLite, REST APIs, Microservices, Docker, AWS EC2

Tools: Git, GitHub Actions, Linux, CI/CD

Concepts: Distributed Systems, Fault Tolerance, OOP, Data Structures & Algorithms, Agile/Scrum